MILITARY SPECIFICATION

REGULATOR, VOLTAGE CN-514( )/GRC

1. SCOPE

* 1.1 This specification covers one type of voltage regulator designated Regulator, Voltage CN-514( )/GRC. It is a 5.0 KVA automatic ac voltage regulator which provides the means of regulating and distributing the primary power source for radio relay equipment. Case, Standardized Components, Electrical CY-2851( )/GRC is a part of Regulator, Voltage CN-514( )/GRC. (See 6.1 and 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

* FEDERAL

QQ-S-781	Steel Strapping, Flat
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated And Solid, Sheet Stock (Container Grade) And Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic (And Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing And Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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* MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus-resistant, Of Communications, Electronic, And Associated Electrical Equipment
MIL-P-11268	Parts, Materials, And Processes Used In Electronic Communication Equipment
MIL-M-11473	Magnetic-particle Inspection, Soundness Requirements For Weldments
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish For Ground Signal Equipment
MIL-S-24251	Shield, Retainer (Bases) And Adaptors, Electron Tube Heat Dissipating, General Specification For
MIL-C-55169	Connector, Electric, Type Q, Multi-contact

* STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-147	Palletized And Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, Or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification Of Visual And Mechanical Defects
MIL-STD-454	Standard General Requirements For Electrical Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-681	Identification Coding And Application Of Hookup And Lead Wire
MIL-STD-810	Environmental Test Methods

* DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List Of Accessories For Package Tester
DL-SM-A-422830	Regulator, Voltage CN-514( )/GRC
SC-DL-423900	Case, Standardized Components, Electrical CY-2851( )/GRC
SC-A-678017	Burn-in/Debugging Procedure For Regulator, Voltage CN-514( )/GRC

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

* 3.1 Preproduction samples. - The contractor shall furnish three preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2(d)(1))

* 3.2 Construction. - The equipment shall be constructed in accordance with the requirements of this specification, and of drawings DL-SM-A-422830 and SC-DL-423900 (including the requirement for parts, materials, and processes thereon). (See 4.14)

* 3.3 Parts, materials, and processes; general. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 6.8)

* 3.4 Welding. - Welding shall be capable of meeting the tests of 4.11. (See 3.20)

3.5 Identification of wiring. - Wiring shall be identified by color-coding or, if approved by the contracting officer, may be identified by other means such as marking of terminals at both ends of each lead. Color-coding for chassis wiring shall be system I or II conforming to MIL-STD-681. (See 3.20)

3.6 Connectors. - Connectors used in the equipment shall be multi-contact, type C, conforming to the requirements of MIL-C-55169. (See 4.4)

* 3.7 Shields. - Electron tube shields used in the equipment shall be of the heat dissipating type and shall conform to the requirements of MIL-S-24251. (See 4.4)

* 3.8 Tubes, Electron. - Electron tubes shall conform to requirement 29 of MIL-STD-454. (See 4.4)

* 3.9 Semiconductor devices. - Semiconductor devices shall conform to requirement 30 of MIL-STD-454. (See 4.4)

* 3.10 Finish. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4)

3.11 Marking. - Marking shall conform to MIL-M-13231. (See 4.4) Front panel marking shall be group I as described in that specification.

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3.11.1 Visibility. - Whenever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.11.2 Serial numbers. - The following item shall be serial numbered:

Regulator, Voltage CN-514( )/GRC

3.12 Electrical requirements. -

* 3.12.1 Input voltage. - The voltage regulator shall provide the output specified when connected to a single phase power source of 95 to 135 vac, 47.5 to 63.5 hz. (See 4.7.1)

3.12.2 Output. -

* 3.12.2.1 Output voltage adjust. - With an input of 115 vac, 47.5 to 63.5 hz, the voltage regulator output shall be capable of adjustment over the range of 110 to 120 vac throughout the load range of 0 to 43 amperes. Final output voltage setting shall be 115 vac. (See 4.7.2.1)

* 3.12.2.2 Automatic operation. - When subjected to the input voltage requirements of 3.12.1 the voltage regulator shall supply at the applicable front panel receptacles and connectors an output voltage of 115 vac (± 1.0%) throughout the load range of 0 to 43 amperes. (See 4.7.2.2)

* 3.12.2.3 Manual operation. - Should failure of the automatic regulation circuitry occur, it shall be possible, by front panel control, to manually adjust the output voltage to 115 vac. (See 4.7.2.3)

* 3.12.2.4 Unregulated output voltage. - Unregulated output supply voltages shall be available at the "Unregulated 115V Utility Output" receptacles on the front panel. (See 4.7.2.4)

* 3.12.3 Distortion. - The total harmonic distortion contributed by the regulator shall not exceed 1.5%. The distortion requirement shall be met over the specified range of line voltage, line frequency and load current. (See 4.7.3)

* 3.12.4 Correction rate and stability. - The equipment output voltage shall correct for voltage changes and load changes at a rate not less than ten (10) volts per second. When subjected to maximum load or line voltage changes, the closed-loop control system of the regulator shall not exhibit any prolonged or continues hunting (oscillation). (See 4.7.4)

3.12.5 Overload Protection. - The input of the regulator shall incorporate a magnetic circuit breaker which will protect the regulator from overload. (See 4.7.5)

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3.13 Service conditions. - The equipment shall meet the following service conditions:

* 3.13.1 Operation (general). - Continuous use for a period of one (1) year, under the environmental requirements specified in 3.13.2.1 through 3.13.2.6, with a duty cycle of eight (8) hours on and sixteen (16) hours off with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.8. Approval of the preproduction sample (See 3.1) shall be considered as compliance with this requirement.

* 3.13.2 Service conditions operation requirements. - Unless otherwise specified, prior to, during, and after testing as required for each of the following service conditions, the equipment shall conform to the requirements of 3.12.2.2, 3.12.2.3, 3.12.3, 3.12.4, and 3.12.5.

3.13.2.1 Low Temperature. - See 4.8.1

3.13.2.2 High temperature. - See 4.8.2

3.13.2.3 Humidity (moisture resistance). - See 4.8.3

3.13.2.4 Altitude (elevation). - See 4.8.4

3.13.2.5 Orientation. - See 4.8.5

3.13.2.6 Bounce (vibration) and shock. - The equipment shall be capable of meeting the requirements of table I.

Table I - Bounce and Shock

Par. Heading	Test Para.	Performance Reqmts
Bounce (vibration)	4.8.6	Specified performance (See 3.13.2 and Note A) No physical damage except minor surface abrasion.
Shock; bench-handling	4.8.7	Specified performance (See 3.13.2) No physical damage
Shock; drop	4.8.8	Operable (See 3.16) Any physical damage shall be minor only.

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measurements may also be required at the other end of the cable.

NOTE A: With the equipment shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 hertz.

* 3.14 Electromagnetic interference characteristics. - The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461, table A-I-MIL-E-55301, Equipment Class CE. Tests and test methods shall be as specified in MIL-STD-462. (See 4.9) If the contractor considers it desirable to modify the equipment proper to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

* 3.15 Overall treatment (moisture and fungus proofing - treatment (MFP). - Moisture and fungus proofing treatment shall conform to MIL-T-152. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.10. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened surfaces.

* 3.16 Operation (final). - When subjected to the test specified in 4.7.c, the equipment shall perform satisfactorily as indicated on the "Regulated Output Voltage" meter. (See 6.7)

* 3.17 Burn-in/Debugging preconditioning. - Each equipment shall meet the requirements herein, without subsequent processing, after subjection to the burn-in cycle of 4.6. (Also see 4.5)

* 3.18 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.12)

3.19 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on order. (See 6.2(g) and 6.2(h))

* 3.20 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268:

(1) General requirements for plastic material and plastic parts.

(2) Wiring and cabling, including: Slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable, identification of wiring.

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(3) Flux and cleaning agents for solder.

(4) Soldering.

(5) Cleaning (equipment).

(6) Riveting.

(7) General requirements for securing of parts.

(8) Welding.

(9) Tropicalization.

(10) Controls.

(11) Self-tapping screws.

(12) Gears.

(b) In this Specification:

(1) Marking.

(2) Finish.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

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(2) Quality conformance inspection of preparation for delivery. (See 4.15)

* 4.3 Preproduction inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and environmental/design (See tables III, IV, and II respectively). The inspection requirements of 4.4, group A, and group B shall be performed on all samples prior to environmental/design inspection. Environmental/design inspection shall be performed on the samples specified and in the sequence shown in table II.

Table II. - Environmental/design inspection

Inspection	Req. Par.	Insp. Par.
SAMPLE NO. 1		
Low temperature	3.13.2.1	4.8.1
High temperature	3.13.2.2	4.8.2
Orientation	3.13.2.5	4.8.5
Humidity (moisture resistance)	3.13.2.3	4.8.3
SAMPLE NO. 2		
Altitude	3.13.2.4	4.8.4
Electromagnetic interference	3.14	4.9
Shock; drop	3.13.2.6	4.8.8
SAMPLE NO. 3		
Shock; bench-handling	3.13.2.6	4.8.7
Bounce (vibration)	3.13.2.6	4.8.6
Fungus	3.15	4.10

* 4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Connectors	3.6
Shields	3.7
Tubes, electron	3.8
Semiconductor devices	3.9
Finish	3.10
Marking	3.11

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* 4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to burn-in/debugging (See 3.17) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4) Test Equipment for Government verification inspection shall be made available by the contractor.

* 4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.7.6) shall be last. (See 6.7)

Table III. - Group A inspection

Inspection	Req.Par.	Insp.Par.	AQL Major	AQL Minor
Visual and Mechanical				
Regulator, Voltage CN-514( )/GRC	3.2, 3.20	4.13	1.0%	4.0%
Case, Standardized Components, Electrical CY-2851( )/GRC	3.2, 3.20	4.13	1.0%	4.0%
Electrical				
Input Voltage	3.12.1	4.7.1	2.5 DPHU* for the group combined	**
Output voltage adjust	3.12.2.1	4.7.2.1		
Automatic operation	3.12.2.2	4.7.2.2		
Manual operation	3.12.2.3	4.7.2.3		
Unregulated output voltage	3.12.2.4	4.7.2.4		
Distortion	3.12.3	4.7.3		
Correction rate and stability	3.12.4	4.7.4		
Operation (final)	3.16	4.7.6	0.1%	**

* DPHU - Defects Per Hundred Units
** All Electrical defects are considered major

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* 4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table IV and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

Table IV - Group B inspection

Inspection	Req. Par.	Insp. Par.	AQL Major	AQL Minor
Interchangeability	3.18	4.12	6.5%	*
Welding	3.4	4.11	4.0%	*
Overload protection	3.12.5	4.7.5	6.5%	*

* All defects are considered major.

* 4.5.2.1 Group B sampling plans. - The group B sampling plans, for the AQL's listed in Table IV, shall be as follows:

AQL	INSPECTION LEVEL
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

* 4.5.3 Group C inspection. - This inspection shall consist of the tests specified in table V and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1.

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Table V - Group C inspection

Inspection	Req. Par.	Insp. Par.
Group I		
Low temperature	3.13.2.1	4.8.1
High temperature	3.13.2.2	4.8.2
Humidity (moisture resistance)	3.13.2.3	4.8.3
Group II		
Altitude	3.13.2.4	4.8.4
Electromagnetic interference	3.14	4.9
Shock, drop	3.13.2.6	4.8.8
Group III		
Bounce (vibration)	3.13.2.6	4.8.6
Fungus	3.15	4.10
Group IV		
Orientation	3.13.2.5	4.8.5
Shock; bench-handling	3.13.2.6	4.8.7
Group V		
Equipment Verification Review	3.2	4.14

4.5.3.1 Sampling for Group C inspection. -

* 4.5.3.1.1 Group I, Group II, and Group III - Table V. - One (1) sample shall be selected at random for each group, from each 400 equipments produced, and subjected to the inspections specified in the order shown. The first samples shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

* 4.5.3.1.2 Group IV - Table V. - One (1) sample shall be selected at random from each 200 equipments produced and subjected to the inspections specified, in the order shown. The firstsample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

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4.5.3.1.3 Group V - Table V. - The Equipment Verification Review (EVR) shall be performed on the initial production lot. No subsequent inspection for EVR will be required. (See 4.14)

* 4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.5)

4.5.3.3 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

* 4.6 Burn-in/Debugging Procedures. - Each equipment shall be subjected to the conditioning period specified in ECOM drawing SC-A-678017. After completion of the conditioning period, the unit shall not be repaired, alined or otherwise changed prior to subjection to acceptance inspection. (See 3.17 and 4.5)

4.7 Electrical tests. -

* 4.7.1 Input voltage. - Regulator, Voltage CN-514( )/GRC shall be operated from a single phase, ac, variable power source to determine compliance with 3.12.1 Input test voltages and frequencies shall be as specified in the applicable test paragraph.

4.7.2 Output. -

* 4.7.2.1 Output voltage adjust. - Regulator, Voltage CN-514( )/GRC shall be tested using the test configuration shown in Figure 1. With the regulator connected to a power source specified in 4.7.1, the output voltage shall be varied (using the "output voltage" adjust control) over the voltage range specified in 3.12.2.1. The closed-loop regulating system shall perform satisfactorily throughout the required output range. Input test voltages, frequencies, and output load requirements to determine conformance to 3.12.2.1 shall conform to table VI.

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Table VI - Test conditions

INPUT		LOAD
Volts, ac	Frequency (hz)	(Amperes)
115	47.5, 50, 60, 63.5	0, 43

* 4.7.2.2 Automatic Operation. - The test configuration shown in Figure 1 shall be used to determine compliance with 3.12.2.2. The "output voltage" adjust control shall not be disturbed after the final setting has been made (See 3.12.2.1) and the test for automatic operation ("Manual-Automatic" Switch in the "Automatic" position) initiated. Input test voltages, frequencies, and output load requirements for this test shall conform to table VII.

Table VII. - Test conditions

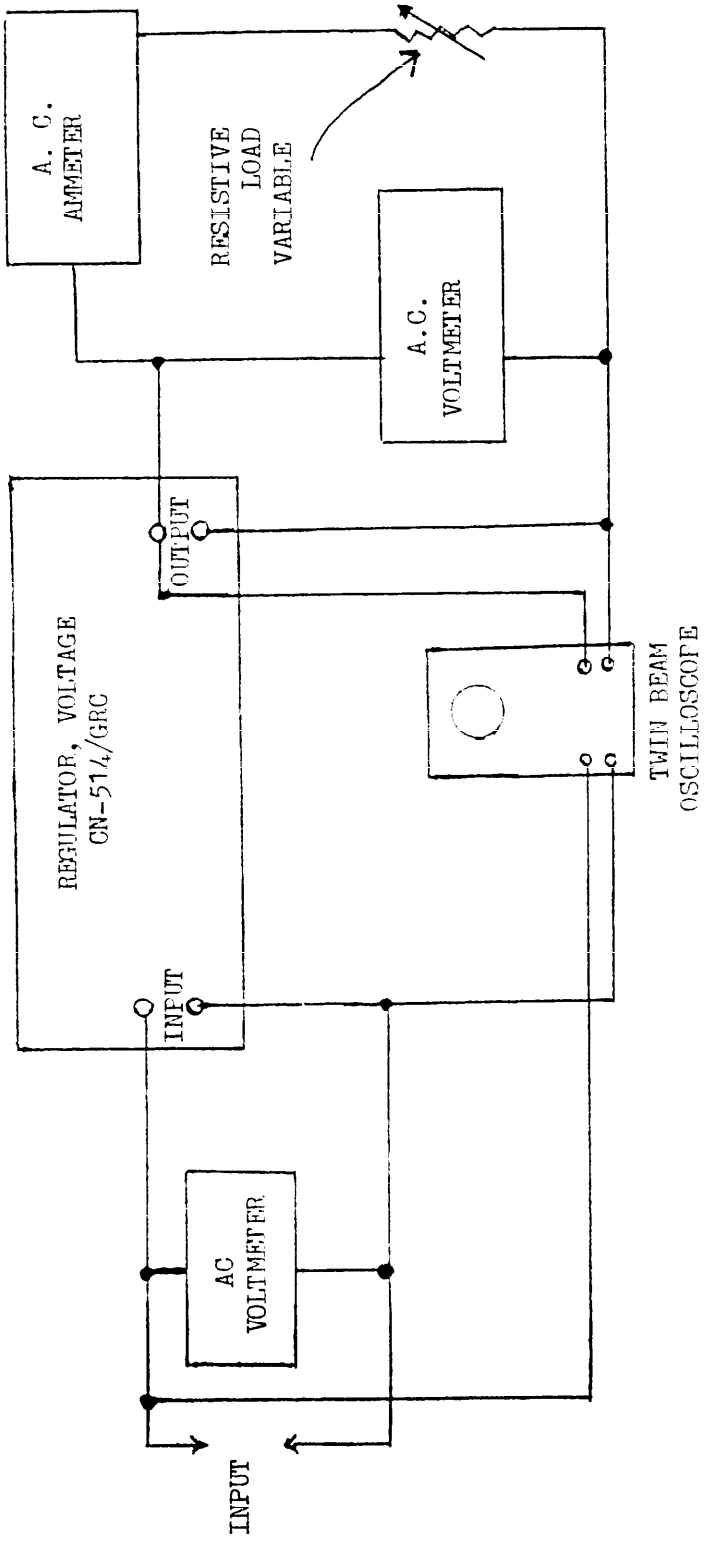
INPUT		LOAD
Volts, ac	Frequency (hz)	(Amperes)
95	47.5, 50, 60, 63.5	0, 43
115	47.5, 50, 60, 63.5	0, 43
135	47.5, 50, 60, 63.5	0, 43

* 4.7.2.3 Manual operation. - The test configuration shown in Figure 1 shall be used to determine compliance with 3.12.2.3. The "Manual-Automatic" switch shall be placed in the "Manual" position; and by operation of the "Raise-Lower" switch, the output voltage shall be capable of adjustment to the output voltage specified. Input test voltages, frequencies, and output load requirements for this test shall conform to table VIII.

Table VIII. - Test Conditions

INPUT		LOAD
Volts, ac	Frequency (hz)	(Amperes)
95	50, 60	0, 43
115	50, 60	0, 43
135	50, 60	0, 43

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TEST CONFIGURATION

FIGURE 1

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* 4.7.2.4 Unregulated output voltage. - The voltage as measured at the unregulated output voltage receptacles shall read the same as the input voltage. (See 3.12.2.4) Input test voltage and frequency shall be 115 vac, 60 hz.

* 4.7.3 Distortion. - The test configuration shown in Figure 1 shall be used to determine compliance with 3.12.3. The input and output waveforms shall be analyzed to determine the magnitude of harmonic distortion. Input test voltages, frequencies, and output load requirements for this test shall conform to table VIII.

* 4.7.4 Correction rate and stability. - The test configuration shown in Figure 1 shall be used to determine compliance with 3.12.4. The test will be performed as follows:

(a) The regulator to be tested shall be energized using an input of 115 vac, 60 hz; and the output set at 115 vac, no load.

(b) Place the "Manual-Automatic" switch on the front panel in the "Manual" position. Adjust the output voltage to 125 vac by activating the "Raise-Lower" switch.

(c) With a suitable timing device, note the time interval for the output voltage to return to 115 volts from 125 volts when the "Manual-Automatic" switch is placed in the "Automatic" position. The time interval shall conform to that specified in 3.12.4.

(d) Place the "Manual-Automatic" switch in the "Manual" position and adjust the output voltage to 135 vac by activating the "Raise-Lower" switch.

(e) Return the "Manual-Automatic" switch to the "Automatic" position. The output voltage shall return to 115 vac without prolonged or continuous hunting (oscillation).

(f) Repeat the above procedure except that the specified output voltage (125 vac) in steps (b) and (c) shall be 105 volts; and the specified output voltage in step (d) shall be 95 volts.

* 4.7.5 Overload protection (See 3.12.5). - The voltage regulator output shall be shorted. Upon removal of the short and resetting of the circuit breaker, the voltage regulator shall resume normal operation. Test conditions: Input - 115 vac, 60 hz, Load - 43 amps.

* 4.7.6 Operation (final) (See 3.16). - Regulator, Voltage CN-514( )/GRC shall be operated from a single phase, 115 vac, 60 hz power source under full load conditions. Both the "Manual" and "Automatic" operating positions will be tested for evidence of satisfactory operation. (See 4.5.1 and 4.7)

* 4.8 Service condition tests. - The following tests shall be performed on Regulator Voltage CN-514( )/GRC. (See 4.8.1 through 4.8.8). Unless otherwise specified, input test voltages, frequencies, and load conditions for all service condition testing shall conform to Table VII.

* 4.8.1 Low temperature (See 3.13.2, 3.13.2.1, and 6.8). - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -80°F for 6 hours. The operating temperature shall be stabilized at -20°F. Prior to testing, the equipment shal meet specification performance for the measurements specified in 3.13.2.

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* 4.8.2 High temperature (See 3.13.2, 3.13.2.2, and 6.8). - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 132°F. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2.

* 4.8.3 Humidity (moisture-resistance) (See 3.13.2, 3.13.2.3, and 6.8). - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If the equipment fails to meet specified performance during cycling it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment fails to meet the specified performance during cycling or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

* 4.8.4 Altitude (elevation) (See 3.13.2, 3.13.2.4, and 6.8). - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810 except that the chamber pressure (step 3) shall be 12,000 ft. above sea level. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2.

* 4.8.5 Orientation (See 3.13.2, 3.13.2.5, and 6.8). - The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 (plus 3, minus 1) degrees. Measurements shall be made during inclination in each plane for specification performance as specified in 3.13.2.

* 4.8.6 Bounce (vibration) test (See 3.13.2 and 3.13.2.6). - The equipment shall be tested in accordance with Method 514, Procedure XI of MIL-STD-810 except that the test sample will not be placed on the "front panel" face at any time during the test period. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439.

* 4.8.7 Shock test, bench-handling (See 3.13.2 and 3.13.2.6). - The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to testing, the equipment shall meet specification requirements for the measurements specified in 3.13.2.

* 4.8.8 Shock test, drop (See 3.13.2 and 3.13.2.6). - The equipment shall be tested in accordance with Method 516, Procedure II, of MIL-STD-810 except that the height of drop shall be 24 inches. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2.

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* 4.9 Electromagnetic interference characteristics. - Conformance of the test sample in accordance with 3.14 shall be determined by the contractor or an approved testing laboratory. Testing shall be in accordance with MIL-STD-462. The test item shall be operated using a 115 vac, 60 hz input. Prior to testing, the contractor shall furnish the contracting officer, for approval, a test plan which shall include a block diagram of the test set-up and a list of the test equipment utilized. Upon approval by the contracting officer, and provided all other requirements of this specification are met, the approved test sample shall be used as a model for visual comparison of production units.

* 4.10 Fungus test (See 3.15). - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

* 4.11 Welding inspection (See 3.4). - Welding shall be inspected for nonvisual defects by the magnetic particle method specified in MIL-M-11473 and by the ultrasonic method using a Reflectoscope, as made by Sperry Products, Incorporated, Danbury, Connecticut, or equal. Both tests shall be performed on each sample unit.

* 4.12 Interchangeability. - The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.18. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers and insertable assemblies; when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating of parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

* 4.13 Visual and mechanical inspection. - The equipment shall be examined for the defects listed in MIL-STD-252. (See 3.2 and 3.20)

* 4.14 Equipment Verification Review. -

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4.14.1 The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings cited for construction (par. 3.2). The EVR shall consist of the following:

4.14.2 An audit to establish that "as-built" equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

4.14.3 An audit to establish that "as-built" configuration of unit, assembly, module, or part meet the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.14.4 The contractor shall compile an EVR report(s) containing the following information:

(a) Identification of unit, assembly, module, or part and details proving compliance with 4.14.2 and 4.14.3 above.

(b) Discrepancies noted.

(c) Corrective action taken.

4.14.5 The contractor shall make available to the Government copies of his EVR report(s) no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

(a) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(b) As required, responsible personnel from each functional department available for discussions in their respective areas.

(c) Adequate administrative support for the EVR.

4.15 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

* 4.16 Rough handling test (preparation for delivery). - When rough handling test in accordance with MIL-P-116 is required by contract (See 6.2(d)(2), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

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Inspection	Insp. Par.	Req. Par.
Operation (final)	4.7.6	3.16

* 5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. - Preservation and packaging shall be level A or C as specified. (See 6.2(b))

5.1.1 Level A.

5.1.1.1 Cleaning. - Each Regulator, Voltage CN-514( )/GRC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each Regulator, Voltage CN-514( )/GRC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged Method IC-1.

5.1.1.4.2 Fuses. - Each fuse shall be packaged in accordance with method III as follows: cushion each fuse by wrapping together in paperboard conforming to PPP-P-291, type 1. Secure wrap with gummed paper tape conforming to PPP-T-45, type III, grade A. Secure the packaged fuses to the control panel of the set with tape conforming to PPP-T-76.

5.1.1.4.3 Lamps. - Each lamp shall be individually packaged method III as follows: cushion each lamp by wrapping in paperboard conforming to PPP-P-291, type 1. Secure wrap with gummed tape conforming to PPP-T-45, type III, grade A. Secure the packaged lamps to the control panel of the set with tape conforming to PPP-T-76.

5.1.1.4.4 Regulator, Voltage CN-514( )/GRC. - Each regulator shall be packaged method III as follows: cushion the regulator on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c, weather resistant. Place the technical literature packaged as specified in 5.1.1.4.1 on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. - Regulator, Voltage CN-514( )/GRC shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited in transit storage.

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5.2 Packing. - Packing shall be level A, B or C as specified.

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of containers packaged as specified in 5.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5c or V3c. The load shall be "banded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirement of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585 style 2 or 3, class 3. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. -

5.2.2.1 Palletized load. - A quantity of containers, packaged as specified in 5.1 shall be packed as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Regulator, Voltage CN-514( )/GRC, packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, style E. The gross weight of the box shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the box specification. Reinforcing shall be by nonmetallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, type II, and PPP-T-97, type IV, respectively. Selection of the material and application shall be in accordance with the appendix of the box specification.

5.2.3 Level C. -

5.2.3.1 Palletized load. - A quantity of Regulator, Voltage CN-514( )/GRC packaged as specified in 5.1 shall be palletized as specified in 5.2.1.1 except that the fiberboard cap shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Regulator, Voltage CN-514( )/GRC, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640, Class 1.

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5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. - Regulator, Voltage CN-514( )/GRC is for use in fixed plant, intermediate or forward areas where the primary voltage source varies from 95 to 135 volts and the frequency varies from 47.5 to 63.5 hertz. It is capable of providing good regulation of the power source and supplying an output voltage of 115 vac. It is used to power radio relay equipment such as the AN/GRC-50(V)( ).

* 6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendments thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(c) The specific paragraphs of Section 5 which are applicable to this procurement.

(d) Preproduction inspection:

(1) Three sample units of each item cited in section 1 are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.1)

(2) Preproduction pack(s) as follows:

Makeup of pack(s).

Number of each kind of pack to be submitted.

Inspection to be performed thereon.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.19)

(h) Quantity of tools and running spare parts required. (See 3.19)

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(i) Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.15)

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CN-514W/GRC. The contractor should apply for nomenclature (Type Designation) in accordance with the applicable clause in the contract. (See 1.1)

6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of the corrective action. (See 4.5.3.2)

* 6.6 Color. - A color chip to match the color specified (See 3.10) may be obtained, if required, upon request to: Commanding General, Coating and Chemical Laboratory, Aberdeen Proving Ground, Maryland.

* 6.7 Location of operational inspection. - It is desirable that the operational inspection (4.7.6) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging. (See 4.5.1)

* 6.8 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, construction methods, etc.

* 6.8.1 Temperature. -

(a) Operating: Ambient temperature in the range of +132°F to -20°F (The high temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature not to exceed 6 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 6 hours, at any one time.

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* 6.8.2 Relative humidity. - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

* 6.8.3 Altitude (elevation). -

(a) Operating: Up to 12,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

* 6.8.4 Orientation. -

(a) Operating.-Any orientation up to 20° from normal operating position (that is: forward, backward, left, or right).

(b) Nonoperating. - Storage in any position for a period of two years.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-55223F(EL) REGULATOR, VOLTAGE SR-5114( )/GRC

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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